COMP 4602

Social Networking

Project Overview

Carleton University

Project Overview

- **Project Overview** posted on Brightspace
  - Found under **Final Project** (scroll down)
  - **Group selection activity** available on Brightspace (under Tools -> Groups, closes Feb 1)
- **Presentations** – Late March / Early April (approx. last 3 weeks)
- **Final Report** due mid April

Topics

- Should relate to **social network theory** (broadly defined)
  - Anything with a significant **network** component or network dataset (does *not* have to be "social")
  - Anything with a significant **graph theory** component
    - If your final project does not have a network that you can point to and talk about, *something has gone wrong*

Topics

- Approaches:
  - **Empirical** (data gathering)
  - **Experimental** (coding / simulations)
  - **Theoretical** (math)

Empirical

- **Application** of social network theory to your choice of domains
  - **Gather data** from your favorite domain
    - Does not need to be world-shaking – it can be *something fun*
    - Ex: analysis of relationships on a season of Survivor
    - Special template for "**interactions in media**" (linked on Brightspace)
    - You do need to establish the **importance** / **generalizability** of the topic
    - Construct a data set from trace data (see A1)
    - Can also be a **pre-made dataset**, but you need more analysis
  - **Analyze** the network using concepts from class
    - What hidden **structures** is there in the network? How does that affect the domain?
    - What **predictions** would the theory make based on the structure? Does this mirror reality?

Experimental

- **Application** of social network theory to experiments / simulations
- Identify a **research question**
  - Build a **simulation**, using some of the theories from class, to try to answer the research question
    - Assumption should be well-founded with appropriate references
    - Network may be synthetic or based on data
  - **Analyze** the data from the simulations
    - What does your simulations say about the **original research domain**?
    - What are your assumptions and **limitations**?

Others

- **Develop** your own ideas
  - Apply concepts from class to **other research projects** you may be involved in
    - Come talk with me if this other project is also "for-credit"
  - Open to **different kinds of projects** not listed here. Talk to me during office hours or at the consultation.
  - Anything **graph theory** / **networks** related is game!

Proposal

- **Proposal** (3%)
  - Useful **links** and **ideas** posted on Brightspace
- Get useful **feedback** about the **feasibility** of your proposed topic, your research plan, and potentially additional ideas related to the topic.
  - What is the **research question** you are trying to answer?
  - What aspect of it **interests** you?
  - How do you **plan** to approach the question?
  - Where do you intend to find **addition resources** / **literature**?

Consultations

- **Project Consultations** before the break
  - Tentatively planned for Feb 12 & 13
  - Each group can meet individually with me
  - Pitch you idea(s) and get some feedback
  - Feel free to book an appointment / come to office hours / email me before that

Proposal

- What is your **topic** and what is the **research question** that you are trying to answer?
- What aspects of the topic **interest** you? How do you plan on exploring that aspect?
  - This may incorporate additional theoretical frameworks or simulations and experiments.
- Where do you intend on **finding resources** that will help you answer your research question? (ex: data)
- What **relevant literature** have you already identified? (esp. for surveys)

Presentation

- **Presentation** (20%) – Late March – Early April
- Introduce about your **research problem** and your **approach** to investigating it
  - **Progress** and interim results
  - Possible **extensions** and invitation for feedback
  - Later presentations should showcase **more results**
- Assessment based on **progress** and **effective communications**

Progress

- When you submit your final project, you must also include a paragraph describing your **progress** since your presentation
  - Highlight how you **incorporated feedback** into the paper
  - For early presentations, this can be a very high level description, since most of the work is likely done after the presentation

Feedback

- **Feedback** (5%) – Early April
- You will be expected to provide **feedback** (short reviews)
  - Will provide a template for this
  - Total number of reviews TBA
- Presenter will receive **anonymized aggregated** feedback

Project

- **Project Report** (32%)
- Follow an "academic research paper" **template** (suggested sections detailed in Overview)
  - Recommended sections are for **theory** / **experimental** papers
  - **Survey** papers mainly group related ideas together
- Report – **10-12 pages** (Word default)
  - You may use 7 pages in the AAMAS LaTeX submission format

References

- **Any reference format** is accepted, so long as information is complete
- You must use **inline references** [1].
  - [1] Inline references appear at the end of sentences or paragraphs where you present **another person's ideas** or results
- Most of the paper should be in **your own words**
- "**Quotations** should be used when you are reusing words. Use these sparingly." [2]

Grievances

- Undergrads will work in **teams of 2-3**
  - Higher quality work expected from teams of 3
  - Graded as a group
  - Assume you be all **contributing equally** to the final product
  - If this is not the case, you can submit a **Grievance** with documentation
  - Goal of the process is to **rectify the situation** – what needs to be done so that everyone contributes equally
    - Grade adjustments and forking the project are also possible outcomes

Important Dates

Dates in this version of the document is **tentative**.

Group Selection	Feb 1
Project Consultation	Week of Feb 12
Proposal Due	Feb 26
Presentations	Last ~3 Weeks of Classes
Final Submission	April 17